



**DELHI PUBLIC SCHOOL
NEWTOWN SESSION:
2020-21
ANNUAL EXAMINATION (ONLINE)**

CLASS: IX

FULL

MARKS: 60

SUBJECT: COMPUTER APPLICATIONS

TIME: 1

Hour 30 minutes

Instructions:

- Section A is compulsory.
- Answer any three questions from Section B.
- This paper consists of four printed pages.
- All working including rough work must be clearly shown on the same sheet as the rest of the answers.
- VDT should be written for every program.

Section A (15 Marks)

[Attempt all questions from this section.]

Question 1

- a) In what way data hiding is related to data abstraction? Discuss with a real life example.

[2] Convert the following if-else-if construct into switch case:

[3]

```
if(var==1)    System.out.println("Distinction");  
else if(var==2)    System.out.println("First Division");  
else if(var==3)    System.out.println("Second Division");  
else    System.out.println("invalid");
```

b)The main function in a Java program is declared as:

[2]

```
public static void main( )
```

How this method is treated differently with other member methods of a class.

What is the significance of the word *void*?

d) What will be the output of the following code:

[2]

```
int k = -1; float j = 2.2f;
```

$$\mathbf{k} / = - \mathbf{k} / ++\mathbf{j} - \mathbf{k}++ \% ++\mathbf{j} ;$$
$$\mathbf{k} = \mathbf{k} / (- \mathbf{k} / ++\mathbf{j} - \mathbf{k}++ \% ++\mathbf{j});$$
$$k = -1/0$$

```
System.out.println(j);
```

System.out.println(k);

e) Predict the output of the following functions:

[2]

i) Math rint(16.5); ii) Math.round(-18.7);

f) Write the equivalent java expression for the following:

$$^{[2]} \quad \frac{1}{2a^{2i}} \left(\frac{\sqrt{b^2 - 4a}}{2a} \right) + y^3$$

g) +00000000000000000000

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**h) Predict output of the following program,
if the method is invoked as *call("A24")*:**

```
[2]    void call(String n)  
{    String st= n + "comp";  
      System.out.println("n= " + (n+32));  
      System.out.println("st= " + 12+st);  }
```

Section B (45 Marks)

[Attempt any three questions from this section.]

The answers in this Section should consist of the Programs in BlueJ environment. Each program should be written using Variable descriptions/Mnemonic Codes so that the logic of the program is clearly depicted. Flow-Charts and Algorithms are not required.]

Question 2

[15]

Write a program to input a number. Count and print the frequency of each digit present in that number. (do not use nested loop to do the program)

The output should be given as:

Sample Input: 44514621

Sample Output:

=====	
Digit	Frequency
=====	
1	2
2	1
4	3
5	1
6	1

Question 3

[15]

Create a class *Tax_calculation* as given

Class Name : *Tax_calculation*

Data Members/ Variables : *name, age, monthly_salary, phone_no, designation*

Member Functions/ Methods:

i. *void accept()* :accepting the required data members.

ii. *double calculate()* :to calculate the income tax of the employee based on the

following condition:

Annual income (in ₹)	Tax
Upto 130000	No tax
Between 130000 – 200000	15% of the amount
From 200000 – 250000	₹ 5000 + 20% of the amount

From 250000- 330000	₹ 10000 + 30% of the amount
Above 330000	₹ 10000 + 35% of the amount

iii. *void print()* : to display all the details of the employee with annual income and income tax in a tabular format.

Create the above class and implement all the functions. Write a *main()* method to show the necessary action.

Question 4

[15]

Design a class Armstrong_series with two methods:

- **boolean getArmstrong(int n)** it checks and returns true if number n is armstrong otherwise returns false. A positive integer of n digits is called an Armstrong number of order n (order is number of digits) if,

$$abcd... = \text{pow}(a,n) + \text{pow}(b,n) + \text{pow}(c,n) + \text{pow}(d,n) +$$

Example: Input : 153

Output : Yes, 153 is an Armstrong number. $(1*1*1 + 5*5*5 + 3*3*3 = 153)$

- **void printSeries()** to generate first ten armstrong numbers by calling getArmstrong(int n) method to determine if the number is armstrong or not.

[Hint: 1,2,3,4,5,6,7,8,9,153]

Question 5

[15]

Write a program to generate a triangle or an inverted triangle based

upon user's choice of triangle to be displayed. (do not use nested loop to do the program)

Example 1:

**Input: Type 1 for a triangle
triangle**

Enter your choice: 1

Sample Output:

**1
2 2
3 3 3
4 4 4 4
5 5 5 5 5**

Example 2:

Input: Type 2 for an inverted

Enter your choice: 2

Sample Output:

**5 5 5 5 5
4 4 4 4
3 3 3
2 2
1**